# Independent reanalysis of the largest sporadic Creutzfeldt–Jakob disease genome-wide association study confirms all published loci and refines the STX6 signal

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## Abstract

**Background.** Sporadic Creutzfeldt–Jakob disease (sCJD) is the most common human prion disease. The largest genome-wide association study (GWAS) of sCJD (GCST90001389; 4,110 cases, 13,569 controls; *Lancet Neurology* 2020) reported three genome-wide significant loci at PRNP, STX6 and GAL3ST1. Independent replication of summary statistics by parties outside the original consortium remains rare.

**Methods.** We downloaded the publicly deposited summary statistics (6,314,492 variants, GRCh37) and performed an end-to-end independent quality control and replication analysis using only open-source tooling implemented from first principles in Python's standard library: genomic-control inflation ($\lambda$), genome-wide significance screening, locus annotation against Ensembl GRCh37 coordinates, Wakefield approximate Bayes factors, linkage-disequilibrium clustering against the 1000 Genomes phase 3 panel via the Ensembl REST API, and stratified $\lambda$ diagnostics by minor-allele-frequency bin.

**Results.** (1) We independently reproduce **all three published loci**: PRNP chr20:4,672,307 ($p = 1.62\times10^{-15}$), GAL3ST1 chr22:30,950,360 ($p = 6.18\times10^{-10}$) and STX6 chr1:180,961,245 ($p = 7.51\times10^{-9}$); 41 variants reach $p < 5\times10^{-8}$; no malformed records were found; global $\lambda = 1.0587$. (2) Fine-mapping with real LD shows the STX6 signal is a single cohesive haplotype block: the lead cluster ($r^2 \geq 0.80$ with anchor rs11586493) carries **90.5% of regional posterior mass**; at PRNP, 100% of posterior falls within a single linkage block around codon 129 (rs1799990 appears among proxies, $r^2 \approx 0.68$); the GAL3ST1 signal is poorly tagged by common-panel proxies (max $r^2 \approx 0.45$), consistent with a lower-frequency haplotype. Notably, **rs3747957 — the index variant independently identified by an Oxford multi-omic study in *Brain* (2025) — is present in the 2020 sumstats with $p = 9.74\times10^{-9}$ and identical effect direction ($\beta = -0.148$)**, ranked 11th regionally, demonstrating that the 2020 data already contained evidence later validated functionally. (3) Stratified $\lambda$ analysis across MAF bins spans only 0.016 across minor-allele-frequency strata (<0.05: $\lambda$=1.062; ≥0.45: $\lambda$=1.070), arguing against substantial residual population stratification. (4) As a parallel verification exercise on expression data, we reproduce exactly the neuronal-loss/gliosis transcriptional signature of GSE160208 (184 differentially expressed genes under the original authors' criteria; r = 1.000 for top-ranked genes), and show that the blood microRNA signature of GSE140069 (Nat Commun 2020) loses formal significance after standard adjustment for age, sex and RIN (84 miRNAs significant at FDR<0.05 without covariate adjustment → 1 after adjustment), although directionality and nominal significance of the four discovery miRNAs persist ($p = 7\times10^{-4}$–$4\times10^{-2}$).

(5) A five-cohort brain eQTL meta-analysis (~2,182 samples) shows the STX6 lead variant is a genome-wide significant brain eQTL ($p = 6.6\times10^{-47}$) with 89% effect-direction concordance and complete regional posterior concentration on both-signal hypotheses (H2+H3+H4 = 1.000 across all parametrizations; H3+H4 ≈ 0.995 under our default prior; the standard R `coloc.abf` implementation with its own prior assigns 98% to H4), supporting expression-mediated mechanism; a splicing (sQTL) scan of the same five cohorts finds **no significant cis-sQTL for STX6 in any of them** (0 clusters; the GWAS signal also does not colocalize with any neighbouring-gene splicing signal), making expression the only detectable molecular trait at the locus; and (6) a blind whole-genome clumping scan rediscovers all three published loci as the top three clusters (3/3), demonstrating pipeline sensitivity rather than circular confirmation.

**Conclusions.** Public prion-disease summary statistics support full independent replication by non-consortium parties. We document a previously unreported consistency between the 2020 sCJD GWAS and the 2025 functional index variant rs3747957, provide cluster-level credible sets for all three loci, and quantify how covariate confounding can inflate blood-based biomarker signatures. All code, checksums and reports are openly available.

# 1. Introduction

Human prion diseases are fatal neurodegenerative conditions caused by misfolding of the cellular prion protein (PrP^Sc). Sporadic CJD accounts for ~85% of cases. Beyond the PRNP codon 129 modifier, host genetic modifiers were established by the 2020 GWAS of Mead et al., which identified PRNP, STX6 and GAL3ST1 at genome-wide significance.

Independent reanalysis serves three purposes that the original study cannot: (i) it verifies that public deposition is complete and internally consistent; (ii) it stress-tests conclusions with methods chosen independently of the original pipeline; (iii) it lowers the barrier for citizen-science participation in rare-disease research, following the precedent of Minikel & Vallabh.

Here we report a fully independent, from-scratch reanalysis of GCST90001389 and two companion expression datasets (GSE160208 brain tissue; GSE140069 whole blood), executed without access to individual-level genotypes and with all statistics implemented from first principles.

# 2. Data

| Resource | Accession | Content |
| --- | --- | --- |
| sCJD GWAS sumstats | GCST90001389 | 6,314,492 variants, GRCh37, β/SE/p/EAF |
| Brain expression | GSE160208 | 47 samples (27 CJD / 20 controls), NanoString 800 genes |
| Blood microRNA | GSE140069 | 57 sCJD / 48 controls, 939 miRNAs |
| LD reference | 1000 Genomes phase 3 (ALL) | via Ensembl REST ld endpoint |

Integrity: SHA-style MD5 checksums recorded before analysis (Appendix of project repository).

# 3. Methods

## *3.1 GWAS quality control*

Streaming parse of the compressed sumstats (single pass, constant memory). Malformed-line count, allele-frequency sanity, $\chi^2$ statistic per variant as $(\beta/SE)^2$, genomic-control factor $\lambda = \text{median}(\chi^2)/0.454936$ computed globally and within pre-registered MAF strata (<0.05; 0.05–0.25; 0.25–0.45; ≥0.45). Genome-wide threshold $p < 5\times10^{-8}$.

## *3.2 Locus annotation*

GRCh37 gene coordinates retrieved live from Ensembl REST (never from memory) for PRNP, STX6, GAL3ST1 windows (±50 kb around lead).

## *3.3 Approximate-Bayes-factor fine-mapping*

Per-variant Wakefield ABF with prior variance W = 0.04 on log(OR): $ABF_i = \sqrt{SE^2/(SE^2+W)}\cdot\exp(\chi^2_i W/(2(SE^2+W)))$. Regional posteriors $\pi_i \propto ABF_i$. Because individual genotypes are unavailable, we report **cluster-level credible mass**: variants are grouped by pairwise $r^2 \geq 0.80$ with the regional lead using Ensembl REST LD (1000G phase 3 ALL); posterior mass of the lead cluster versus remainder quantifies whether the signal is one haplotype block or dispersed. This is explicitly an approximation — joint models (SuSiE/FINEMAP) require genotypes.

## *3.4 Expression analyses*

GSE160208: Welch t-tests on log2-transformed normalized intensities, frontal-cortex-only contrasts, Benjamini–Hochberg FDR; exact replication of the original authors' criteria ($p<0.05 \wedge |\log_2 FC|>1$) alongside our FDR<0.05 criterion. GSE140069: log2(x+1) transform; primary model = OLS log2 ~ group + sex + age + RIN (covariates from the series matrix; the original publication adjusted age); sensitivity analyses unadjusted and detection-filtered. Effect sizes as Cohen's d. All statistics cross-validated against R anchors (t distribution CDF and BH procedure matched to ≤10^-13 relative error; permutation calibration of FDR).

### *3.5 Colocalization with brain eQTLs*

For the STX6 region (chr1:180.9–181.1 Mb GRCh37), we tested whether the GWAS signal and STX6 expression quantitative trait loci (eQTLs) share a causal variant, using the regional approximate-Bayes-factor colocalization framework (Giambartolomei et al. 2014) implemented from the published equations: prior W = 0.04^2; priors p1 = p2 = 10^-4, p12 = 10^-5 (sensitivity: 10× more conservative). eQTL summary statistics were retrieved as remote tabix region queries (pysam/htslib) from the eQTL Catalogue release 8, which uniformly reprocessed GTEx v10 and independent brain cohorts: CommonMind DLPFC (n=586), ROSMAP DLPFC (n=560), BrainSeq DLPFC (n=479), GTEx v10 DLPFC (n=285) and GTEx v10 cerebellum (n=272). eQTL coordinates (GRCh38) were converted to GRCh37 with a single Ensembl-mapped offset (–30,864 bp), and alleles harmonised to the ALT/effect convention (0 incompatible variants discarded). Because no single cohort is individually powered, we additionally performed an inverse-variance-weighted meta-eQTL across the five cohorts (~2,182 brain samples), yielding a lead-cluster eQTL signal of p = 6.6×10^-47. Implementation validation: (i) an independent reimplementation of the ABF/posterior equations in R agrees with our Python code to 6 decimal places (H3 = 0.994997 vs 0.9950); (ii) the standard R `coloc.abf` (coloc 5.2.3) reproduces the combined both-signal posterior (H2+H3+H4 = 1.000) with per-variant PP.H4 Spearman ρ = 0.91 against our ABF-diagonal mass — while demonstrating that the shared-vs-distinct split (H4 vs H3/H2) is prior-dependent, as declared. LD-panel sensitivity: re-running the fine-mapping with the population-matched 1000 Genomes EUR panel leaves the STX6 lead-cluster posterior unchanged (90.5% at r^2≥0.8), improves PRNP tagging (58.9% → 80.9%), and leaves GAL3ST1 poorly tagged in both panels (consistent with a lower-frequency haplotype). Splicing scan: leafcutter sQTL summary statistics for the same five cohorts (eQTL Catalogue r8 `cc` files = significance-passed cis pairs per intron cluster; cluster identity normalised by genomic coordinates) were harmonised to the GWAS and tested by colocalization per cohort×cluster; clusters were attributed to genes by the catalogue's gene_id column (STX6 = ENSG00000135823, GRCh37 chr1:180,941,861–180,992,047, verified via Ensembl REST lookup).

### *3.6 Blind whole-genome clumping (discovery sensitivity)*

To demonstrate that our pipeline would find the known loci rather than merely confirm them, we scanned all 6,314,492 variants blind (threshold p < 10^-5) and applied greedy distance clumping (±500 kb; lead = smallest p within cluster), with no locus annotation used until after cluster definition. Known loci were matched post hoc.

## 4. Results

### *4.1 Sumstats integrity and inflation*

Zero malformed records among 6,314,492 lines. Global λ_GC = 1.0587 (liminal, acceptable). Stratified λ:

| MAF stratum | n | λ |
|---|---|---|
| <0.05 | 325,236 | 1.0617 |
| 0.05–0.25 | 532,664 | 1.0579 |
| 0.25–0.45 | 328,920 | 1.0547 |
| ≥0.45 | 76,078 | 1.0703 |

Gradient 0.0156 — inconsistent with major residual population stratification, which preferentially inflates common variants.

### *4.2 Independent replication of all three loci*

| Locus | Our best hit (GRCh37) | p | β | Annotation |
|---|---|---|---|---|
| PRNP | chr20:4,672,307 C>T | 1.62×10^-15 | –0.219 | PRNP region; rs60704301/rs2093390/rs4254562 |
| GAL3ST1 | chr22:30,950,360 T>C | 6.18×10^-10 | –0.169 | GAL3ST1 promoter/5′ region |
| STX6 | chr1:180,961,245 G>A | 7.51×10^-9 | –0.149 | intragenic STX6 (Ensembl 180,941,861–180,992,047) |

All 41 genome-wide-significant variants fall within these three regions; none elsewhere. A blind whole-genome clumping scan (Section 3.6) independently rediscovers all three loci as the top three clusters, confirming pipeline sensitivity rather than circular confirmation.

### 4.3 Cluster-level fine-mapping

- **PRNP**: 337 regional variants; anchor rs60704301 (merged → rs2093390). Posterior mass 100% within r^2≥0.50 of the anchor in **both LD panels** (58.9% at r^2≥0.80 with the ALL panel; **80.9%** with the population-matched EUR panel); codon-129 variant rs1799990 appears among proxy rsIDs. The entire signal is one haplotype structure around the prion-protein gene. - **STX6**: 162 variants; anchor = lead rs11586493; 20/20 top variants panel-covered, max r^2 = 1.00; **90.5%** of mass in the r^2≥0.80 lead cluster — a single cohesive block that includes rs3747957 (r^2 = 0.99); **identical (90.5%) under the EUR panel**. - **GAL3ST1**: 322 variants; lead absent from 1000G phase 3 (anchor rs386462923→rs8142452, rank 3); only 4/20 top variants panel-covered; best cross-r^2 ≈ 0.45 — **unchanged under the EUR panel**. The signal is poorly tagged by common proxies — consistent with a lower-frequency haplotype and an explicit caveat for imputation-based replication.

### 4.4 Consistency with the 2025 functional index variant

The Oxford multi-omic study (*Brain*, 2025) nominated synonymous/missense variant **rs3747957** (chr1:180,953,853 GRCh37) as the STX6 index. In the 2020 sumstats this variant has p = 9.74×10^-9, β = –0.148 — same direction as our lead (β = –0.149) — ranking 11th of 162 regional variants. The 2020 dataset therefore already carried the association later validated functionally; we supply the explicit numerical bridge.

### 4.5 Expression signatures replicate exactly (brain) but are fragile under covariate adjustment (blood)

Brain GSE160208: 437/800 genes FDR<0.05; 184 DEGs under original criteria — identical count and rank order (r = 1.000 top-10). Blood GSE140069: unadjusted log2-Welch yields 84 significant miRNAs (10↑/74↓); OLS adjusting age+sex+RIN leaves **1** (hsa-miR-500a-3p); the four discovery miRNAs retain direction and nominal significance (p = 7×10^-4–4×10^-2) but only hsa-miR-93-5p survives FDR within the detection-filtered universe (q = 0.048). Cases were on average 12.8 years older than controls (66.4 vs 53.6) with lower RNA integrity (RIN 5.59 vs 6.50) — a textbook confounding structure that the original paper partially addressed (age via Partek GSA) and that fully explains the discrepancy between naive and adjusted counts. Furthermore, a novel cross-compartment integration analysis (not performed by either original study) shows that experimentally validated targets of the four discovery blood miRNAs are NOT over-represented among the up-regulated brain DEGs (hypergeometric, all q = 1.0; overlaps at or below chance), arguing that the blood signature reflects peripheral processes rather than the cerebral transcriptional program — cautioning against naive blood→brain causal inference.

### 4.7 Colocalization: the STX6 GWAS signal sits inside a significant brain eQTL block

Per-cohort colocalization is power-limited (eQTL n ≤ 586; the GWAS lead is a suggestive single-cohort eQTL with concordant direction in 78–81% of regional variants). The five-cohort meta-eQTL, however, renders the lead GWAS variant a **genome-wide significant STX6 eQTL (p = 6.6×10^-47; z ≈ 14)** with **89% effect-direction concordance** across 390 harmonised variants. In the regional colocalization framework, H0+H1 collapse to ≈0 and the entire posterior sits on both-signal hypotheses — **H2+H3+H4 = 1.000 in every parametrization tested** (H3+H4 ≈ 0.995 under our default prior; the standard R `coloc.abf` 5.2.3 with its own prior assigns 98% to H4): both signals are real and confined to the same LD block. Our implementation was validated by an exact reimplementation of the ABF/posterior equations in R (agreement to 6 decimal places) and by the standard `coloc.abf` package (per-variant PP.H4 Spearman ρ = 0.91 against our ABF-diagonal mass) — which also demonstrates that the shared-vs-distinct split (H4 vs H3/H2) is prior-dependent under r^2 ≥ 0.97 LD, as declared. The strict split between H4 (shared causal variant) and H3 (distinct causal variants) is formally unidentifiable here because all cluster members are in r^2 ≥ 0.97 — a documented limitation of colocalization under strong LD. Given (i) the significant meta-eQTL at the GWAS lead, (ii) direction concordance, and (iii) our fine-mapping showing a single posterior cluster, the totality of evidence supports regulation of STX6 expression as a plausible mechanism of the association, consistent with the independent functional nomination of rs3747957 by the Oxford multi-omic study.

### *4.8 Splicing: no significant cis-sQTL for STX6 in any brain cohort*

The eQTL Catalogue release 8 provides leafcutter splicing-QTL summary statistics for the same five brain cohorts. We scanned every significant intron-cluster (the `cc` files contain only significance-passed cis pairs) whose cis window intersects the STX6 region, harmonised variants to the GWAS and ran the validated colocalization per cohort×cluster (6 tests total). **Zero significant cis-sQTL clusters for STX6 exist in any cohort** — in contrast to the strong expression QTL (p = 6.6×10^-47) — and no neighbouring-gene splicing signal (KIAA1614, QSOX1, RP5-1180C10.2) colocalizes with the GWAS either (all PP.H4 ≈ 0; Bonferroni-aware interpretation). At this locus, expression is the only detectable molecular trait; a splicing-mediated component is neither supported nor formally excluded (absence of evidence in significance-filtered files is not evidence of absence), but there is no positive splicing signal to interpret.

### *4.9 Blind clumping rediscovers all published loci*

The blind scan produced 35 independent clusters; the top three are exactly PRNP (chr20:4,672,307; p = 1.62×10^-15), GAL3ST1 (chr22:30,950,360; p = 6.18×10^-10) and STX6 (chr1:180,961,245; p = 7.51×10^-9) — **3/3 known loci rediscovered without any locus hint**, in correct significance rank. The strongest residual cluster (chr16:15,539,902; p = 5.73×10^-8) is borderline genome-wide significant and merits independent follow-up; we report it as an observation, not a novel-locus claim.

### *4.10 Ethics and scope*

No individual-level human data were generated or obtained beyond public deposits; no patient-identifiable information was processed. This work is a verification contribution and makes no clinical claims.

## 5. Discussion

Our results deliver the three things an independent reanalysis can uniquely provide. First, **verification**: every number in the deposited sumstats parsed cleanly, all 41 genome-wide-significant variants sit exactly where the original consortium reported them, and genomic inflation is liminal and uniform across allele-frequency strata — the public record of the largest sCJD GWAS is trustworthy, and we publish the checksums and code to let anyone re-verify this in minutes.

Second, **the STX6 association is expression-mediated to the limit that summary statistics allow**. A five-cohort brain eQTL meta-analysis makes the GWAS lead variant a genome-wide significant eQTL of STX6 (p = 6.6×10^-47) with concordant effect direction in 89% of regional variants, and regional colocalization concentrates the entire posterior on shared-block hypotheses. Strict single-variant attribution (H4) is unidentifiable under r^2 ≥ 0.97 — we say so explicitly — but the combination of significant meta-eQTL, direction concordance and a single fine-mapping cluster is exactly the pattern expected if the 2020 association and the

2025 functional nomination of rs3747957 are two views of the same regulatory event.

Third, **a previously unreported numerical bridge between two landmark studies**. The 2025 Oxford multi-omic study functionally nominated rs3747957 as the STX6 index variant. We show the 2020 sumstats already contained that association at $p = 9.74\times10^{-9}$ with matching effect direction — evidence that was present but unremarked for five years. This is a concrete demonstration of why open summary statistics matter: discoveries sometimes sleep in deposited data until someone looks.

Fourth, **a cautionary quantification for biomarker research**. The blood microRNA signature for sCJD (Nat Commun 2020) collapses from 84 FDR<0.05-significant miRNAs (no covariate adjustment) to 1 after standard covariate adjustment, because cases were on average 12.8 years older than controls. Directionality and nominal significance of the four discovery miRNAs survive — consistent with a genuine but weaker-than-presented signal. We stress this is not an accusation: the original authors adjusted age themselves; our contribution is making the magnitude of naive-vs-adjusted divergence explicit and reproducible for future biomarker pipelines.

Methodologically, we demonstrate that a complete GWAS quality-control, fine-mapping and colocalization pipeline can run with a standard-library statistical core (all inferential statistics implemented from first principles and cross-validated against R anchors), relying on external open-source libraries only for figure rendering (matplotlib) and remote tabix access to eQTL summary statistics (pysam/htslib). This lowers the entry barrier for researchers in resource-limited settings, including in countries like Brazil where sCJD surveillance exists but prion-genetics capacity is thin.

Finally, our cluster-level credible analysis shows the STX6 signal is one cohesive LD block whose posterior mass concentrates on the lead haplotype rather than dispersing across independent false positives. Formal joint fine-mapping with individual genotypes remains the gold standard; we offer ours as the honest ceiling achievable from summary statistics alone.

## 6. Limitations

- Summary-statistics-only: no conditional/joint fine-mapping; cluster-level inference only. - Expression datasets lack individual-level covariates for GSE160208 (no age/PMI metadata). - Our OLS implementation cannot reproduce Partek's gene-specific variance correction; differences in Section 4.5 may partly reflect estimator choice. - Colocalization under $r^2 \geq 0.97$ cannot split H3 (distinct variants) from H4 (shared); we report block-level support plus direction concordance instead of an H4 point estimate. - Brain eQTL cohorts (DLPFC/cerebellum, adult) only approximate the affected tissue and cell types in sCJD; cell-type-specific (e.g. microglial) eQTL may differ. - Single-author independent initiative; peer review pending (bioRxiv DOI upon submission).

## 7. Data & code availability

Sumstats GCST90001389 (GWAS Catalog); GEO GSE160208/GSE140069. Full pipeline (pure-Python stdlib), reports with every intermediate number, figure-generation scripts and MD5 checksums: github.com/BlackYuriJDU/dcj-lito (release v1.0.0-preprint; archived on Zenodo, DOI 10.5281/zenodo.22164910).

## 8. References

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*Figures: volcano_gse160208.png, volcano_gse140069.png, heatmap_top_genes.png (pipeline/reports/figuras/). Fine-mapping numbers from relatorio_finemap_loci.md v2 and relatorio_lambda_gc.md. All statistics cross-checked against R anchors; adversarial audit report included in the repository* *(colaboracao/laudo_estatistico_adversarial.md).*